Section A — Quantitative Aptitude

1. Divisibility: What is the smallest number that must be added to 1,789 to make it divisible by 11?

2. Number Formation: Find the smallest four-digit number greater than 2,000 that ends in 5 and is divisible by 9.

3. Averages: The average score of 15 students increases by 3 marks when one student's score of 37 is replaced. Find the new student's score.

4. Pipes and Cisterns: Pipe A fills a tank in 6 hours, while pipe B empties it in 8 hours. They are opened alternately for one hour each, starting with A. How long will the empty tank take to fill?

5. Simple Interest: A sum becomes 1.5 times its original value in 8 years at simple interest. Find the annual interest rate.

6. Work and Production: Three machines produce 240 components in 4 hours. At the same rate, how many components can five machines produce in 6 hours?

7. Probability: A fair die is rolled twice. What is the probability of obtaining at least one six?

8. Letter Series: Find the next letter: B, E, J, Q, ?

Section B — Logical Reasoning

9. Ranking: In a class, Priya ranks 12th from the top and 19th from the bottom. How many students are in the class?

10. Coding and Decoding: In a code, each letter is replaced by the letter three positions after it in the alphabet, with the alphabet wrapping around after Z. How is SMART written?

Section C — Data Interpretation

A company recruits candidates from three colleges:

- College A: 120 applicants, 72 selected
- College B: 150 applicants, 105 selected
- College C: 180 applicants, 117 selected

12. Which college has the highest selection percentage?

13. What is the overall selection percentage across all three colleges?

Section A — Quantitative Aptitude

1. Ratio: If 40% of A equals two-thirds of B, find A.

2. Arithmetic Progression: The 7th term of an arithmetic progression is 31, and the 13th term is 55. Find the first term and common difference.

3. Averages: The average of five numbers is 24. Four of the numbers are 18, 20, 25, and 27. Find the fifth number.

4. Clock Accuracy: A clock gains five minutes every hour. It is set correctly at 8:00 AM. What time will it show at 2:00 PM on the same day?

5. Compound Interest: Find the compound interest on ₹10,000 for six months at 20% per annum, compounded quarterly.

6. Races: In a 100 m race, A beats B by 20 m. In another 100 m race, B beats C by 25 m. By how many metres does A beat C in a 100 m race?

7. Probability: Two cards are drawn simultaneously from a standard 52-card deck. Find the probability that both are aces.

8. Number of Factors: How many positive factors does 720 have?

Section B — Logical Reasoning

9. Ordering: Four tasks P, Q, R, and S must be completed subject to these rules: P before Q, R before P, and Q before S. What is the only valid order?

Section C — Data Interpretation

A company's monthly sales are:

- January: ₹2.4 lakh
- February: ₹3 lakh
- March: ₹3.6 lakh
- April: ₹4.2 lakh

10. Calculate the percentage increase in sales from January to April.

11. Calculate the average monthly sales for these four months.

Section A — Quantitative Aptitude

1. Work and Wages: A, B, and C complete a job and receive ₹18,000. A completes half the work, while B and C complete the remaining half equally. How much should each receive if payment is proportional to work done?

2. Number System: Find the smallest positive integer that leaves a remainder of 2 when divided by 5, a remainder of 3 when divided by 7, and a remainder of 4 when divided by 9.

3. Probability: A bag contains four white and six black balls. Two balls are drawn with replacement. Find the probability that the first ball is white and the second is black.

4. Permutations: How many distinct arrangements of the letters in the word LEVEL begin and end with L?

5. Partnership: A invests ₹40,000 for 12 months, and B invests ₹60,000 after four months for the remaining eight months. If the total profit is ₹26,000, find each person's share.

6. Work Rates: Twelve identical pumps can empty a reservoir in 15 hours. How long will 20 identical pumps take to empty the same reservoir?

7. Simplification: Evaluate (999^2-998\times1000) without expanding both large products.

Section B — Logical Reasoning

8. Number Pattern: Find the next number: 1, 2, 6, 24, 120, ?

9. Scheduling: Five presentations A, B, C, D, and E must be scheduled. A must occur before C; B must occur after D; and E must occur before A. Give one valid order satisfying all three conditions.

10. Data Sufficiency: What is the value of integer n?

Statement I: n is a positive integer divisible by 6.

Statement II: n is less than 10 and is divisible by 4.

Determine whether either statement alone, or both together, are sufficient to determine.

Section C — Data Interpretation

A training centre records the following assessment scores:

- Student A: 72
- Student B: 84
- Student C: 66
- Student D: 90
- Student E: 78

11. Find the median score.

12. If every score is increased by 10% and then reduced by 10%, what is the new score of Student D?

Quantitative Aptitude

1. Profit and Loss

A shopkeeper marks an article 40% above its cost price and gives successive discounts of 10% and 15%. Find the final profit percentage.

2. Time, Speed and Distance

A train 180 metres long crosses a pole in 12 seconds. How long will it take to cross a 420-metre platform at the same speed?

3. Time and Work

A can complete a job in 12 days and B in 18 days. They work together for 3 days, after which A leaves. How many additional days will B need to finish the remaining work?

4. Number System

Find the smallest positive integer that leaves remainder 5 when divided by 8 and remainder 9 when divided by 12.

5. Simple and Compound Interest

The difference between compound interest and simple interest for 2 years at 10% per annum is ₹50. Find the principal.

Probability and Permutations

6. Probability

Two fair dice are rolled simultaneously. What is the probability that their sum is 9 or more?

7. Permutation and Combination

How many three-digit numbers divisible by 5 can be formed using the digits 1, 2, 3, 4 and 5 without repeating any digit?

Logical Reasoning and Mathematics

8. Number Series

Find the next term: 3, 7, 15, 31, 63, ?

11. Logical Reasoning

Eight people attend a meeting. Each person shakes hands exactly once with every other person. How many handshakes occur in total?

12. Clocks

Find the smaller angle between the hour hand and the minute hand at 4:20.

13. Pipes and Cisterns

Pipe A fills a tank in 10 hours, pipe B fills it in 15 hours, and a leak empties a full tank in 30 hours. If all three operate together, how long will it take to fill the empty tank?

Data Interpretation and Analytical Reasoning

Use the following sales data for Q14:

Quarter	Sales
Q1	₹4.8 lakh
Q2	₹6 lakh
Q3	₹5.4 lakh
Q4	₹7.8 lakh

14. Data Interpretation

Calculate the percentage increase in sales from Q1 to Q4.

15. Average

A candidate scores 72, 84, 68, 76 and 90 in five tests. What score must the candidate obtain in the sixth test to achieve an average of 80 across all six tests?